SCHENKTRONICS

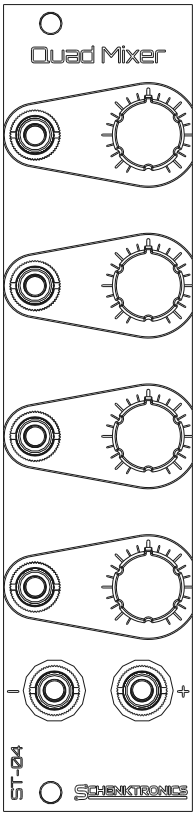
ST-04 Quad Mixer

User Manual

For PCB v1.0 - Updated 2026-10-17

Official version: github.com/Schenktronics/ST04-quad-mixer

Overview



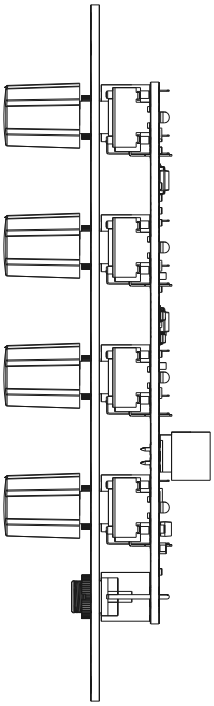
The Quad Mixer is a 6HP mixer with four channels. Each channel has an input jack and a knob, and the knob sets how much of that input goes into the mix, from silent up to its own level. The four channels are summed into two outputs: **+**, the mix as it came in, and **-**, the same mix, inverted.

An input with nothing plugged into it is silent, so only the channels you patch affect the mix.

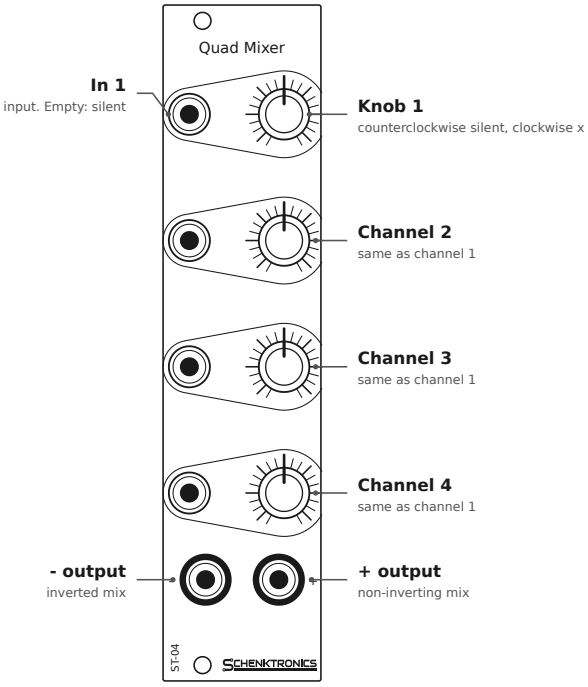
It's an active module, so it needs ±12 V from your case's power supply.

Specifications

Format	Eurorack, 6HP
Panel	30.48 × 128.5 mm
Depth	20 mm
Power	Up to 20 mA on +12 V, up to 20 mA on -12 V, no +5 V. 10-pin header.
Protection	Reverse-power diodes and resettable fuses on both rails, and clamp diodes on each op-amp input
Jacks	6 × 3.5 mm mono (TS): 4 inputs, 2 outputs
Controls	4 x knobs, 100 kΩ linear potentiometers, 6 mm D shaft
Channel gain	0 fully counterclockwise, x0.5 at the center, x1 fully clockwise
Input impedance	About 100 kΩ. A source with a high output impedance loses a little level into it.
Empty inputs	Silent
Outputs	+ (non-inverting mix) and - (inverted mix)
Output impedance	1 kΩ
Output swing	About ±10 V, limited by the op-amps running on ±12 V
Signals	Audio, CV, and gates. DC-coupled.



Panel layout



The panel has four identical channels, numbered 1 to 4 from the top. Each channel has its input jack on the left and its knob on the right. The two outputs are at the bottom: **-** on the left and **+** on the right.

Using the mixer

Knobs

Each knob sets the level its channel adds to the mix:

Knob position	Channel level
Fully counterclockwise	Silent (0)
Center	About x0.5: half the input
Fully clockwise	About x1: the input at its own level

Turn a channel up to add that input to the mix. Turn it fully counterclockwise to leave the input out without unplugging it.

Empty inputs are silent

An input with nothing plugged in is connected to ground, so it adds nothing to the mix, whatever its knob is set to. You can leave unused channels patched or unpatched, and nothing changes in the mix until you plug something in.

The + and - outputs

Both outputs carry the same mix. The **+** output has the same polarity as the inputs. The **-** output is inverted: a rising signal falls, and a +3 V CV becomes -3 V. Use the - output when you want an inverted copy of a mix, or when the module you're patching into needs the opposite direction. The outputs are independent, so you can use either one or both.

Mixing levels

The mix is a sum, so signals add up. Each channel goes up to x1, and the outputs clip above about ±10 V. In practice:

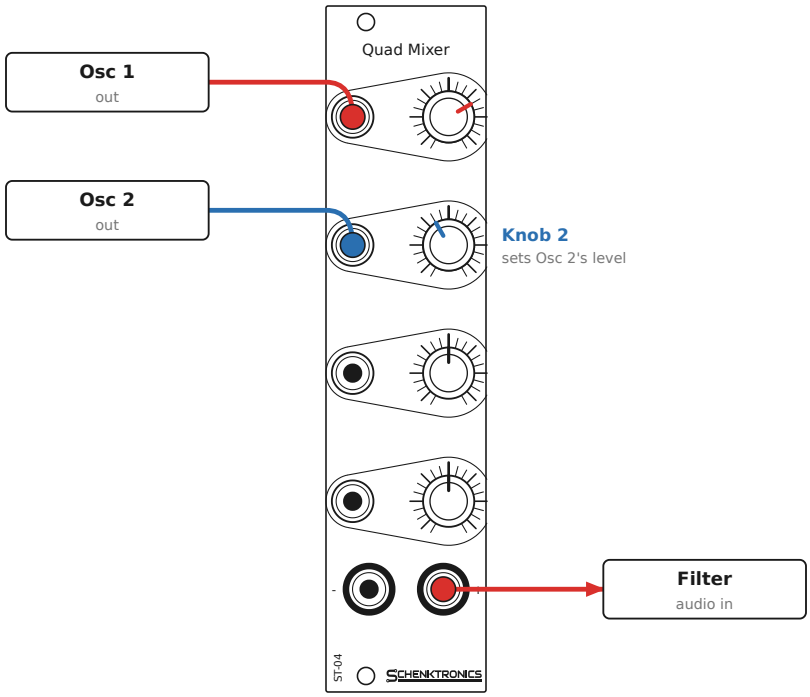
- A single channel at full level clips with an input of about ±10 V. Keep the knob below full for hotter signals.
- Several channels together clip sooner. The outputs clip when the inputs, added together, reach about ±10 V with the knobs at full. Four channels at ±2.5 V each, all at full, get there.

If a CV stops tracking or the output is distorting when you don't want it to, turn the loudest channel down.

Patch examples

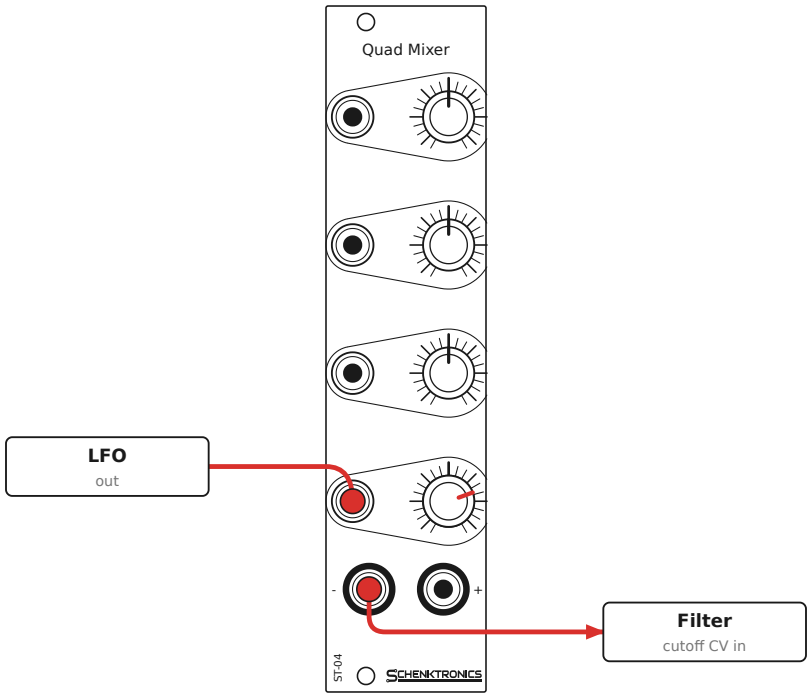
Each drawing shows the modules sending signals on the left and the modules receiving them on the right.

Blend two oscillators



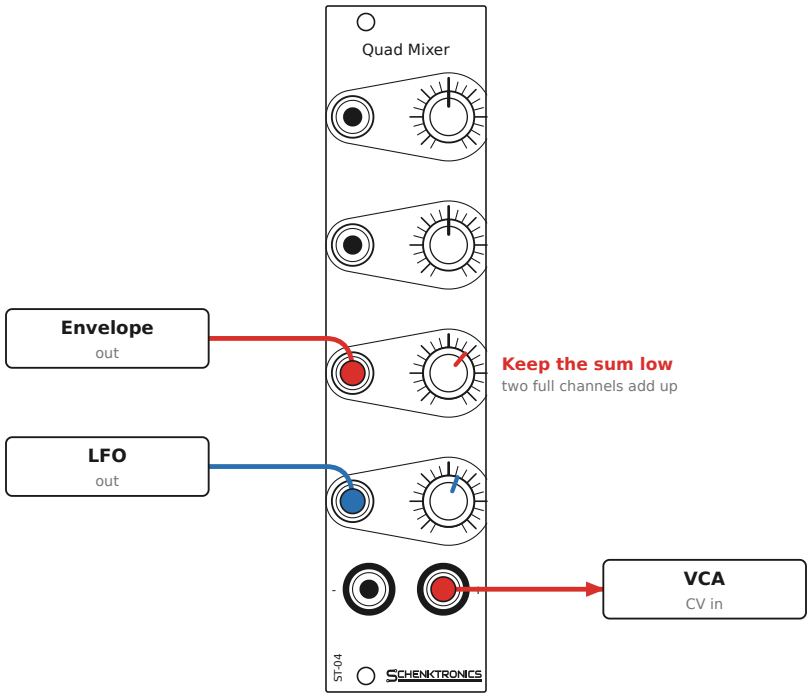
Plug each oscillator into its own channel, and turn the knobs to balance them. Send the + output to the filter, so it gets both oscillators together.

Use the inverted output



Patch an LFO into In 4 and send the **-** output to the filter's cutoff CV. Now the LFO's rise closes the filter instead of opening it. Set the depth with Knob 4.

Sum two CVs



Patch an envelope and an LFO into two channels, and send the **+** output to a VCA's CV input. Both modulate the VCA at once. Keep the knobs below full, since two channels at full can clip the sum.

Good practice

- **Don't patch outputs together.** Joining an output to another module's output makes the two fight. Use another channel of the mixer to combine them instead.
- **Clipping is a level problem.** If the mix clips, or a CV stops at about ±10 V, turn down the knobs or the source levels.
- **Check an unexpected silence.** An empty input is silent, so if a channel you expected to hear is quiet, check that its jack is plugged in and its knob isn't fully counterclockwise.

Installation

1. Turn your case off.
2. Connect a 10-pin to 16-pin Eurorack power cable to the header on the back of the module. The red stripe goes to **-12 V**, marked "Red Stripe" and "-12V" on the board. At the bus board end, the red stripe also goes to -12 V, which is usually marked on the bus.
3. Put the module in any free 6HP space.
4. Fix it in place with two M3 rack screws. Don't overtighten them.
5. Turn the case on. With nothing plugged in, both outputs should read 0 V, whatever the knobs are set to.

The module is protected against a cable plugged in backwards, but don't rely on it. Always check the red stripe before you power on.

Circuit

The inputs are switched jacks, so an empty jack connects its input to ground. A plugged-in signal goes to one end of a 100 kΩ potentiometer, whose other end is grounded. The wiper goes through a 1 kΩ resistor to the non-inverting input of one channel of a TL074 quad op-amp (U1), wired as a unity-gain buffer. Each op-amp input has two clamp diodes, which hold it between about +12 V and -12 V.

The four channels feed a summing amplifier (U2, a TL072) through 10 kΩ resistors. U2's first half inverts the sum, which is the **-** output, through 1 kΩ. Its second half inverts that again, giving the **+** output, through 1 kΩ. A 10 pF capacitor across each of U2's feedback resistors keeps the summing stages stable. Each rail has two 0.1 µF capacitors and a 10 µF capacitor to ground.

The power input has a resettable fuse and a reverse-biased diode on each rail. If the power cable is plugged in backwards, the diodes conduct and the fuses trip, protecting the circuit. The schematic is in hardware/mixer/4ch Mixer.kicad_sch (https://github.com/Schenktronics/ST04-quad-mixer/blob/main/hardware/mixer/4ch%20Mixer.kicad_sch) (KiCad 10).